Inserting a short chord at the same position preserves Knuth equivalence of matchings

Avichai Marmor

doi:10.5802/alco.351; CC BY 4.0

Editorial note. The counted statement is the author's labelled lemma environment at publisher-source.tex lines 1167-1170; its proof environment opens at line 1172, the line after the blank separator, so the pairing is by adjacency and the retained proof range 1172-1176 is the whole environment.

Editorial note. The lemma lem:insert i equivalent insert j (statement, source lines 1157-1160) is retained as context because the counted proof cites it at source line 1173; it is the only outside label of the retained proof.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label lem:insert same chord to equivalent remains equivalent.

Editorial scope. One original result of the pinned publisher article, counted once: invariance of Knuth equivalence under inserting a short chord at a fixed position (author label lem:insert same chord to equivalent remains equivalent).

Allowed standard prerequisites (editorial).

- Insertion of a short chord. insrt_i(m) is the matching obtained by inserting a short chord joining i and i+1 and shifting the later vertices by two.. It is the operation applied to both matchings in the statement.
- Knuth equivalence generated by elementary transformations. Two matchings are Knuth equivalent when one is obtained from the other by a sequence of elementary Knuth transformations.. It is both the hypothesis and the conclusion, and the proof inducts on the length of such a sequence.
- Independence of the insertion position. Inserting a short chord at any two positions gives Knuth equivalent matchings.. It lets the proof restrict to the last position.

Lemma 0.1. Let $m \in \mathcal{M}_N$ be a matching and let $1 \leq i, j \leq N+1$ be indices. Then $\mathrm{insert}_i(m)$ is Knuth equivalent to $\mathrm{insert}_j(m)$.

Lemma 0.2. Let $m_1, m_2 \in \mathcal{M}_N$ be Knuth equivalent matchings, and let $1 \leq i \leq N+1$ be an index. Then $\mathrm{insert}_i(m_1)$ and $\mathrm{insert}_i(m_2)$ are Knuth equivalent.

Proof. By Lemma 0.1, we may assume that $i = N+1$. Assume that $m_2 = \varphi_1 \cdots \varphi_\ell(m_1)$ for some elementary Knuth transformations $\varphi_1, \ldots, \varphi_\ell$. We prove the statement by induction on ℓ. The statement is obvious for $\ell = 0$.

Next, assume that $\ell > 0$. By the induction hypothesis, we may assume that $\mathrm{insert}_{N+1}(m_2)$ is equivalent to $\mathrm{insert}_{N+1}(\varphi_\ell(m_1))$. Therefore, it suffices to show that

$\mathrm{insert}_{N+1}(m_1)$ is equivalent to $\mathrm{insert}_{N+1}(\varphi_\ell(m_1))$. Assume that $(j, j+1),\ (j+2, j') \in m_1$ for some $1 \leq j, j' \leq N$, and that $(j, j'),\ (j+1, j+2) \in \varphi_\ell(m_1)$, as other types of elementary Knuth transformations are handled similarly. Notice that $\mathrm{insert}_{N+1}(m_1)$ and $\mathrm{insert}_{N+1}(\varphi_\ell(m_1))$ have all but four chords in common. Specifically, $\mathrm{insert}_{N+1}(m_1)$ contains the chords $(j, j+1)$ and $(j+2, j')$, while $\mathrm{insert}_{N+1}(\varphi_\ell(m_1))$ contains the chords (j, j') and $(j+1, j+2)$. Therefore, there exists an elementary Knuth transformation φ, such that $\varphi(\mathrm{insert}_{N+1}(m_1)) = \mathrm{insert}_{N+1}(\varphi_\ell(m_1))$. Thus, the matchings $\mathrm{insert}_{N+1}(m_1)$ and $\mathrm{insert}_{N+1}(\varphi_\ell(m_1))$ are equivalent, as required. □